

PS War Room

A ten-factor teardown of the **three software problem statements** shortlisted earlier — scored the way a judging panel actually scores, not the way a pitch deck wants to be read.

Track: Software **Statements reviewed:** 3
Method: PS text + live web research (competitors, papers, regulation)

AT A GLANCE

PS	CORE ASK	BIGGEST RISK	DATA REALITY	VERDICT
● SIH26034	Scan & auto-check packaged-goods labels against Legal Metrology Rules 2011	Font-size-in-mm compliance can't be verified from an uncalibrated photo	Self-collectable — no ministry dataset dependency	● Green
● SIH26107	Source-cited AI assistant for BIS standards & certification guidance	Full IS standard text is paywalled — corpus is inherently partial	Public scheme pages yes, full standards no	● Yellow
● SIH26045	Multilingual RAG assistant across 6+ Ayurveda IP & regulatory regimes	TKDL is access-restricted; six-regime breadth invites shallow answers	India Code / IP India public; TKDL effectively closed	● Yellow

SIH26034 · Software · Agriculture, FoodTech & Rural Development

Compliance checker for packaged commodities under the Legal Metrology (Packaged Commodities) Rules, 2011

Owner: Ministry of Consumer Affairs, Food & Public Distribution **Deadline:** 20 Sept 2026

1 Pain Points & Core Understanding 🔍

Exact problem: build software that scans a packaged commodity's label, image, or e-commerce listing and automatically checks it against the Legal Metrology (Packaged Commodities) Rules, 2011 — mandatory declarations like manufacturer/packer/importer name & address, net quantity, MRP, month/year of manufacture, and consumer-care details, plus correct format, placement, and readability.

Why the problem exists:

- India has millions of SKUs moving through retail, supermarkets and e-commerce — manual inspection by Legal Metrology Officers (LMOs) simply cannot scale to that volume.
- Declaration formatting varies wildly brand-to-brand, so "compliant" isn't a simple keyword check — it's a layout-and-content judgment.
- Records today are kept "physically or through isolated local systems" (the government's own framing when it launched the eMaap portal) — there's no digital trail to cross-check against.
- E-commerce listings are a newer, largely unchecked surface entirely separate from the physical label.

Primary stakeholders: State Legal Metrology Departments & LMOs (enforcement), the Department of Consumer Affairs (policy owner), consumers (harmed by short-weighting or inflated MRP), manufacturers/packers and brand-compliance teams (want to self-check pre-launch), and e-commerce platforms — newly on the hook for origin disclosure under the 2026 amendment below.

Current inefficiencies: inspection is visual and paperwork-heavy, non-compliance (missing declarations, wrong font size, mis-stated MRP) is "frequently observed" per the PS's own background text, and there is no automated first-pass filter to focus scarce officer time on likely violations.

2 Feasibility of Execution

Prototype in hackathon time? Yes — high feasibility. The core loop (image → OCR/field extraction → rule engine → report) is a well-trodden ML pattern; grocery-label OCR is an active, published research area (see §4).

Technical requirements

- OCR / structured extraction: Tesseract, PaddleOCR or EasyOCR for text, or a document-understanding model (LayoutLM/Donut-style) for field-level extraction that's more robust than raw OCR + regex.
- Text-region detection for locating declaration blocks on curved or irregular packaging (bottles, pouches, sachets).
- A versioned rules engine encoding Rule 6 of the 2011 Rules *and* the 2026 amendment (country-of-origin for e-commerce).
- Web/mobile capture app + officer dashboard backend.

Real blockers

- **Font-size-in-millimetres** is genuinely hard to verify from an arbitrary photo without a calibration reference (known object or fixed camera distance) — this is the single hardest literal requirement in the PS.
- No public labeled dataset of "compliant vs. non-compliant" Indian packaging exists — teams build their own ground truth.
- Curved/3D packaging surfaces (not flat labels) break naive OCR pipelines.

Realistic MVP: photo/listing screenshot → extract MRP, net quantity, mfg/import date, manufacturer address, consumer-care text → present/absent/malformed checklist + violation dashboard. Skip true mm-level font verification for the demo and say so — use relative-readability heuristics (contrast, blur, text-height ratio) instead, framed honestly as a scoped limitation rather than hidden.

3 Impact & Relevance 🌍

- **Beneficiaries:** consumers (pricing/quantity transparency), LMOs (far higher inspection throughput), MSMEs (a pre-market self-check tool that avoids penalties), e-commerce platforms (compliance-as-a-service ahead of a real 2026 deadline).
- **Real-world impact:** reduces consumer fraud (short-weighting, backdated manufacture dates, inflated MRP) and formalises fair trade — and it plugs directly into a *live, dated* regulatory mandate.
- **Scalability:** naturally state → national. The new [eMaap national portal](#) is the obvious integration point: eMaap has the *who* (licensed stakeholders), this PS builds the *what* (automated detection).

Why evaluators care: the Ministry of Consumer Affairs owns this PS, so a live demo that visibly catches a real violation on a real product is about as persuasive as an SIH demo gets — it's concrete, not speculative.

4 Scope of Innovation 💡

Competitor / existing-solution landscape

PLAYER	WHAT IT ACTUALLY DOES	GAP VS. THIS PS
eMaap (govt.)	National registration/licensing portal for metrology stakeholders	No image-based label detection at all
Yuka / Open Food Facts	Barcode scan → nutrition/health score	Consumer health scoring, not legal-declaration compliance; not India-specific
ArtworkFlow & similar SaaS	Pre-print packaging workflow tools for brand teams	Internal design review, not enforcement-facing, not automated visual detection
Legal Metrology consultancies	Human advisory (CliniExperts, LegalMetrologyIndia, etc.)	No software at all — manual, paid consulting

Research grounding

- OCR + ML label-transparency pipelines are proven: *Automating Nutritional Claim Verification* (IEEE) and *Tesseract OCR + bounding box extraction on nutrition labels* demonstrate the extraction pattern works.
- A 2025 arXiv study, *Evaluating OCR performance on food packaging labels*, is a useful real-world accuracy baseline to cite — it shows OCR-on-packaging is imperfect in practice, which is exactly the honest caveat judges respect.
- Expiry/manufacture-date detection is its own sub-field (multiple 2018-2024 papers, including a 2024 LLM-assisted approach for accessibility use-cases) — directly reusable for the mfg-date field.

What would make this stand out

- Field-extraction via a document-understanding model rather than brittle OCR+regex — more robust to layout variance.
- A **data-driven rules engine** (JSON rule objects) so the 2026 country-of-origin amendment is a new rule entry, not a rewrite — the single most convincing "we understand this domain" signal.

- A browser-extension mode that runs the same checker against *live e-commerce listings* — directly answers the 2026 amendment and is the most judge-visible differentiator.
- Confidence-scored extraction: flag low-confidence fields for human review instead of silently guessing.

5 Clarity of the Problem Statement 🧩

What's actually asked: web/mobile app, automated extraction + validation, rule-based compliance checking, PDF/editable reports, an officer dashboard, searchable scan history, role-based access.

Where teams misread it: building a generic "OCR text extractor" and stopping there — without encoding the actual Rules as checkable logic, there's no compliance *tool*, just a scanner. The second common miss: treating this as a consumer nutrition-style app when the PS's deliverable list is explicitly *enforcement-first* (dashboards, inspection history, evidence attachment, role-based auth).

Framing that lands: pitch it as enforcement infrastructure, not a scanner app. Lead with the LMO/inspector workflow and audit trail; show the consumer-facing angle as a bonus feature, matching what the PS actually lists as the expected solution.

6 Evaluator's Perspective 🎯

A Consumer Affairs domain official is a plausible panelist, and they will check rule-accuracy, not just UI polish.

- **Matters most:** correctness of the encoded rule text, extraction performance on real crumpled/curved packaging (not one perfect photo), completeness of the officer workflow, and whether the team engaged with the live 2026 e-commerce amendment at all.
- **Red flags:** toy regex checks ("does the string MRP appear"), a demo that only ever works on one staged photo, silence on the font-size-in-mm limitation, no e-commerce angle despite it being current and topical.

7 Team Fit & Execution Strategy 👤

Skills needed: CV/OCR engineer(s), backend/rules-engine developer, frontend/dashboard developer, and — unusually important here — someone who actually reads the legal rule text and turns it into checkable logic.

Ideal 6-person ratio: 2 AI/CV · 2 full-stack · 1 domain/rules-research · 1 UX/presentation. The domain-research seat matters more on this PS than most, because the entire pitch rests on "did you get the rule right."

1. Read the actual Rules 2011 text + the 2026 amendment before writing code; turn every declaration requirement into a checklist.
2. Photograph 30–50 real product labels yourselves (compliant and deliberately flawed) — your own ground truth.
3. Prototype extraction accuracy on that set before committing to architecture.
4. Build the rules engine as data (JSON), not hardcoded conditionals, so it visibly extends on stage.
5. Build the officer dashboard last, once extraction + rules are solid.

8 AI-Buildability Split (20/80) 🤖

The fast 20%: OCR/extraction pipeline wiring, a CRUD dashboard, PDF report generation, basic auth — boilerplate any AI coding assistant handles in hours.

The real 80%: translating legal rule text (including the new 2026 clause) into unambiguous, checkable logic; handling real packaging variance (curves, glare, multilingual text, torn labels) so extraction is trustworthy rather than a demo illusion; deciding what happens on low-confidence extraction; and designing the officer evidence/audit-trail workflow around how enforcement actually happens on the ground.

Risk of leaning on AI alone: an AI-assistant-authored "rule checker" tends to become a shallow keyword matcher — it passes/fails on string presence, not genuine field validation. That collapses the moment a judge tests it with an ambiguous or partially-compliant label, which domain-aware judges specifically try.

On-the-spot test: "Add a check for the 2026 country-of-origin e-commerce rule, right now." If the rules engine is genuinely data-driven, this is a five-minute live edit. If it's hardcoded, the team can't do it — and that gap is obvious immediately.

9 Data & Resource Availability 📊

Reality check: there is no official dataset here — the PS's own "dataset" pointer is the Legal Metrology Act/Rules text itself, a legal document, not training data. This is a from-scratch-data PS, which is actually a strength: no dependency on a ministry that may never hand over data.

- **Public:** the rule text, freely available.
- **Missing:** any public labeled dataset of Indian packaged-commodity labels with compliance annotations.
- **Backup plan:** a two-tier approach — (1) programmatically generate synthetic label mockups with known compliant/non-compliant fields to pressure-test the rules engine deterministically, and (2) layer in self-photographed real products for the OCR-robustness story. This is a strong, honest answer to "what's your dataset."

10 Judge Q&A Stress-Test

Q1 · Your demo photo is clean and flat. What happens on a crumpled, curved, real-world label?

ANSWER Show the confidence-scored extraction — low-confidence fields are flagged for human review instead of silently passed. Cite the real accuracy number from your own self-collected test set, not a guess.

FOLLOW-UP *"What's your actual field-level accuracy on that set?" — have a real, tested number ready.*

Q2 · The Rules require font size in millimetres. How does a photo alone verify that?

ANSWER Be candid: true mm-level verification needs a calibrated capture (reference object or fixed distance), out of scope for a photo-only MVP. The system flags relative-readability risk instead, and is architected so a calibrated-capture mode can be added later without touching the rules engine.

FOLLOW-UP *"So can an LMO issue a penalty off this today?" — answer: flag-for-review today, calibrated verification is the clearly-scoped next step.*

Q3 · How is this different from eMaap, which government is already building?

ANSWER eMaap is a registration/licensing backbone — it doesn't do image-based declaration detection. Position this as the missing enforcement layer sitting on top of eMaap's stakeholder data, not a competitor to it.

FOLLOW-UP *"Have you actually checked eMaap's data model for integration feasibility?" — give an honest answer on what's publicly known vs. assumed.*

Q4 · The 2026 e-commerce amendment doesn't fully take effect until 2027. Why build for it now?

ANSWER Because the rules engine is data-driven, not hardcoded — adding a not-yet-fully-enforced rule costs nothing today and proves the system is built for a moving regulatory target, which is the actual condition Legal Metrology operates under.

Q5 · What stops someone from uploading a fake or edited label to game your checker?

ANSWER For self-check use it's not a security boundary — it's a compliance-assist tool, not a legal certification. For the officer-facing flow, every scan pairs with photo evidence + timestamp/location metadata on the inspection record; tampering is a chain-of-custody problem the workflow addresses, not the OCR model's job.

FOLLOW-UP *"Is that metadata cryptographically signed?" — honest answer: app-side for MVP; hash-signing evidence is a natural hardening step for v2.*

● GREEN LIGHT

Self-collectable data, a live 2026 regulatory hook, and an honestly scopeable failure mode (font-size-in-mm) — green holds only if the team encodes the real rule text instead of shipping a generic OCR demo.

AI-powered intelligent assistant for Indian Standards & BIS services

Owner: Bureau of Indian Standards (BIS) **Deadline:** 20 Sept 2026

1 Pain Points & Core Understanding 🔍

Exact problem: a conversational AI assistant that answers questions about Indian Standards (IS numbers), recommends which standards apply to a given product, explains BIS certification schemes (ISI mark, CRS, hallmarking), guides licensing procedures, points to testing labs, and answers consumer queries — all source-cited and multilingual.

Why it exists: BIS publishes thousands of standards across scattered documents and portals; MSMEs, startups, students and consumers each need a different depth of answer from the same underlying corpus, and today that means manually trawling "multiple documents, portals, and PDFs" — the PS's own words.

Stakeholders: MSMEs/startups seeking certification, consumers wanting to verify what a mark actually means or file a complaint, BIS itself (reducing support-desk load), students/researchers.

Current inefficiency, confirmed directly: the existing [BIS Care app](#) handles ISI/CRS/hallmark verification, complaint filing and recall info — but has no chatbot, no standards-search, and no step-by-step certification guidance. That gap is real, documented, and exactly what this PS asks a team to fill.

2 Feasibility of Execution

Buildable in hackathon time? Yes — this is a textbook retrieval-augmented generation (RAG) build, one of the most mature GenAI patterns available, with strong published precedent (see §4).

Technical requirements

- A corpus from BIS's public portal (scheme pages, FAQs, standard titles/scopes).
- Embedding model + vector store (FAISS/Chroma), an LLM for grounded generation with clause-level citation.
- Multilingual support — not a stretch feature: BIS Care itself already ships in 12 languages, so that's the expected bar.

The real blocker: the full text of most Indian Standards is *paid* BIS publication. A team can index abstracts, scope statements and certification-scheme text, but claiming a "full IS document corpus" would be both legally shaky and practically false.

Realistic MVP: a RAG assistant over BIS's public content that answers "which standards likely apply to my product," walks through the ISI/CRS/hallmarking process end-to-end, and cites its source page every time — explicitly not claiming to reproduce paywalled clause text.

3 Impact & Relevance

- **Beneficiaries:** MSMEs/startups (faster certification cycles), consumers (mark literacy, fewer counterfeit victims), BIS (lower support load).
- **Real-world impact:** mostly economic/efficiency — small manufacturers reach compliance faster — plus consumer-protection value from accurate mark literacy.
- **Scalability:** naturally national from day one — BIS is a single national body (unlike Legal Metrology's partly state-administered structure), and this could plug straight into BIS Care as a new tab.

Why evaluators care: BIS's own app documentation confirms this exact capability is missing today — a working demo addresses a named, verifiable gap rather than a hypothetical one.

4 Scope of Innovation 💡

Competitor / existing-solution landscape

PLAYER	WHAT IT DOES	GAP VS. THIS PS
BIS Care (official app)	ISI/CRS/hallmark verification, complaint filing, recall info	No chatbot, no search, no certification guidance — confirmed gap
Generic LLM chat (ChatGPT etc.)	General-knowledge answers about BIS in plain terms	No guaranteed grounding — will invent IS numbers with full confidence
UMANG / MyGov-style bots	Broad multi-department government query handling	Not standards-depth; too generalist for clause-level accuracy

Research grounding

- [LawPal](#) (2025) — an India-specific legal RAG system (DeepSeek-R1 embeddings + FAISS) reporting over 90% "legal accuracy," though the authors themselves flag that BLEU/ROUGE-based evaluation is limited for legal correctness — a useful architecture template *and* an honest caution about over-trusting your own eval metrics.
- A published RAG chatbot for [Air Force policy/logistics compliance](#) shows the same pattern works for dense regulatory corpora outside the legal domain proper.
- [LexRAG](#) (ACM SIGIR 2025) is a benchmark specifically for multi-turn legal-consultation RAG — citing it signals the team knows evaluating this class of system is itself an open research question.

What would make this stand out

- Hybrid retrieval (keyword + semantic) — standard numbers/codes are exact-match-sensitive in a way pure embedding search handles poorly.
- A guided-intake flow ("which standard applies to my product") for MSMEs who don't know what to ask, rather than open-ended chat only.
- A visible "source last verified/updated" indicator on every citation.

5 Clarity of the Problem Statement 🧩

What's asked: answer IS-standard questions, recommend applicable standards from a product description, explain certification/licensing/testing-lab routes, answer consumer queries, multilingual.

Where teams misread it: wrapping a generic LLM chat with a "you are a BIS expert" prompt and calling it done. Without real retrieval grounding and visible citations, the PS's explicit requirement — "source-backed information... with references to the documents or clauses" — isn't met, and a hallucinated IS number in front of a BIS panelist is fatal to credibility.

Framing that lands: lead the demo with "recommend the applicable standard for my product" (the concretely useful MSME path), keep open Q&A as a secondary mode, and show a citation on *every single answer*, without exception.

6 Evaluator's Perspective 🎯

A BIS official is a plausible panelist and will likely test the assistant with a real, un-rehearsed product description.

- **Matters most:** citation discipline (does it ever answer without a source?), correctness on a spot-check, genuine multilingual quality (correct BIS terminology in-language, not just translation), honest handling of the paywalled-text limitation.
- **Red flags:** no visible citations, confident answers on out-of-scope or fabricated products, a demo that only survives 2–3 rehearsed examples.

7 Team Fit & Execution Strategy

Skills needed: NLP/LLM engineer(s) for the RAG pipeline, backend/API developer, frontend/chat-UI developer, a domain researcher who reads actual BIS certification documentation to validate answers, presenter.

Ideal 6-person ratio: 2 AI/NLP · 2 full-stack · 1 domain-research/QA · 1 UX/presentation.

1. Scope exactly what's legally/practically indexable (public pages, not paywalled standard text) before building anything.
2. Hand-build a 20–30-question gold-answer test set from real BIS documentation — measure retrieval quality, don't eyeball it.
3. Build the guided-intake "which standard applies" flow first — it's the flagship demo path.
4. Add citation UI and confidence-based refusal.
5. Multilingual last, once English retrieval quality is solid.

8 AI-Buildability Split (20/80) 🤖

The fast 20%: the RAG plumbing itself — chunking, embeddings, vector store, LLM call, chat UI — near-boilerplate today, with LawPal's own published architecture serving as a usable template.

The real 80%: scoping a legally-safe corpus (an AI assistant won't flag the paid-standards licensing nuance unprompted); building and grading a real gold-answer evaluation set so "accuracy" is measured, not vibes; designing confidence-based refusal (when to say "I don't know" instead of guessing an IS number — a product-judgment call with real consequences); and building the guided-intake flow, which requires actually understanding how BIS certification categorisation works.

Risk of leaning on AI alone:	a naively-wrapped chatbot will confidently invent plausible IS numbers and certification steps. In a domain where the exact standard number	<i>is</i> the entire point, a hallucinated citation is worse than no answer — and a BIS-aligned judge will catch it on the first real query.
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On-the-spot test: "Ask it about a certification scheme you haven't indexed." Real refusal logic says "I don't have verified information on that scheme." A team that hardcoded a fixed scheme list into the prompt instead of building genuine retrieval will either break or silently hallucinate — visible immediately.

9 Data & Resource Availability 📊

- **Public:** BIS portal pages, scheme descriptions, standard titles/scopes — scrapeable.
- **Paid/restricted:** full IS standard body text, sold via BIS's own e-standard store. No open API exists for BIS content as of this research.
- **If the ideal source is unavailable:** don't claim full-text clause retrieval — the honest scope is "public scheme/process guidance + standard identification," stated up front rather than discovered by a judge's question.
- **Backup plan:** build the corpus from BIS's public pages plus a manually-curated set of 20–30 common consumer-product standard summaries (titles/scope, publicly available), clearly labeled team-curated vs. official-source.

10 Judge Q&A Stress-Test

Q1 · Do you have the actual text of IS standards indexed, or just descriptions?

ANSWER Be upfront — full IS text is a paid BIS publication, so the corpus is built from public scheme/process pages and standard titles/scopes. The system is architected to ingest licensed full text if BIS provided it, but the MVP is honestly scoped to public content.

FOLLOW-UP *"So how do you recommend the right standard without the full text?" — category/scope-level matching is enough to point someone to the correct IS number and process; the deep clause-level question is then correctly redirected to BIS/the paid standard itself.*

Q2 · Show me it answering something you didn't rehearse.

ANSWER This is exactly why confidence-based refusal matters — walk through a genuine low-confidence case live, showing the system say "I'm not certain" rather than avoiding the question.

Q3 · How is this different from just asking ChatGPT?

ANSWER Every answer is grounded with a citation to an actual BIS source page/clause, retrieval is scoped to a curated India-BIS corpus, and the system refuses rather than hallucinates when it lacks a grounded source — three properties generic chat doesn't guarantee.

Q4 · BIS already built BIS Care. Why would they replace it with a student project?

ANSWER Not a replacement pitch — BIS Care's own documentation shows no search/chat/guidance layer exists. This is a natural feature extension that could sit inside the existing app, and the demo should say so explicitly rather than ignore BIS Care.

Q5 · What happens when a standard is revised or withdrawn — does your assistant know?

ANSWER Acknowledge it as a real MVP gap. Show a "source last verified" timestamp as the honest interim answer, and describe — without overclaiming it's built — how a production version would sync against BIS revision notices.

 YELLOW LIGHT

The RAG pattern is proven and BIS Care's own docs confirm the gap is real — but the paywalled-standards constraint means the team must under-promise on corpus completeness, or a domain-expert judge catches it on the first query.

IP-SAKTI Sahayak — multilingual, source-cited AI assistant for Ayurveda IP & regulatory guidance

Owner: Ministry of Ayush **Deadline:** 20 Sept 2026

1 Pain Points & Core Understanding 🔍

Exact problem: a multilingual, RAG-based, source-cited assistant spanning patents, geographical indications (GI), trademarks, copyright, designs, trade secrets, plant-variety rights, Access-and-Benefit-Sharing (ABS) duties, *and* drug-regulatory classification (is a formulation a classical medicine, proprietary medicine, new drug, phytopharmaceutical, food, or cosmetic?) — across national and international regimes.

Why it exists: Ayurveda IP genuinely spans multiple overlapping legal regimes at once — unlike most product categories that sit in a single regime. Practitioners, AYUSH startups, MSMEs and cultivators aren't lawyers and can't be expected to navigate patent law + GI law + biodiversity ABS duties + drug classification simultaneously. Codified and community-held traditional knowledge (TK) further complicates novelty/prior-art questions.

Stakeholders: AYUSH startups/MSMEs commercialising formulations, individual practitioners/researchers, cultivators/TK holders whose community knowledge needs ABS protection, patent examiners (who use TKDL defensively), the Ministry of Ayush as policy owner.

Current inefficiency: per the PS's own background, this is a *twofold* problem — legitimate innovators struggle to navigate overlapping regimes, and scattered documentation raises real TK-misappropriation risk — because no single guidance layer spans TKDL, India Code and IP India's separate portals.

2 Feasibility of Execution

Buildable in hackathon time? Moderately. The RAG pattern itself is proven (identical foundation to SIH26107), but the domain breadth — six-plus overlapping legal regimes plus drug classification — is significantly wider than a single-standards-body assistant. Scope discipline is the entire game here.

Technical requirements

- Multilingual RAG stack (embeddings + vector DB + LLM), same foundation as SIH26107.
- A curated corpus from the sources the PS names: TKDL, India Code, IP India's patent/trademark/design/GI databases.
- Ideally, a **two-stage design**: first classify which legal regime(s) a query touches, then retrieve within that regime — not flat RAG across everything at once.

The real blocker: TKDL has historically operated as a restricted, access-agreement-based defensive tool for patent offices — not an open corpus a hackathon team can freely bulk-download, despite being named as a suggested source in the PS itself. India Code and IP India's databases are genuinely public but fragmented and not built for programmatic bulk access.

Realistic MVP: scope down hard — cover 2–3 of the six-plus named regimes (e.g. patent + GI + drug-classification, the three a real AYUSH startup hits most often), build over India Code + IP India public data + the new AYUSH examination guidelines (see §4), and state TKDL access as a named, known limitation rather than quietly work around it.

3 Impact & Relevance 🌍

- **Beneficiaries:** AYUSH startups/MSMEs (cheaper, faster IP navigation than specialist counsel), TK-holder communities (ABS-obligation awareness reduces unwitting non-compliance), the broader Ayurveda commercialisation ecosystem.
- **Real-world impact:** economic (lower IP-compliance cost for a sector India is actively promoting) and cultural/heritage (helps protect traditional knowledge from misappropriation — TKDL's own founding mission).
- **Scalability:** national by design, and the underlying "RAG over fragmented legal sources" pattern is directly reusable for Siddha and Unani medicine too.

Why evaluators care: this sits inside a live policy moment — the Indian Patent Office recently issued new draft guidelines for examining AYUSH-related inventions. A team that reasons about those guidelines shows current, real research, not a generic idea.

4 Scope of Innovation

Competitor / existing-solution landscape

PLAYER	WHAT IT DOES	GAP VS. THIS PS
TKDL (CSIR)	Defensive prior-art database, mainly for patent examiners	Not a public guidance assistant; restricted access
India Code / IP India portals	Raw legal-document and registry repositories	No synthesis or plain-language guidance layer
LawPal (general legal RAG)	India-focused legal RAG precedent	General legal domain — not Ayurveda-IP-specific, no drug-classification layer

There is **no existing public tool** that does what this PS asks — making it the most greenfield of the three statements. Greenfield cuts both ways: maximum differentiation, but no existing UX pattern to borrow from.

Research grounding

- A CNLU legal-academic paper, *Protecting AYUSH As A Traditional Knowledge In The AI Era*, is directly on-topic and recent — strong evidence the problem is recognised in current literature, not invented for this PS.
- The same RAG-for-regulation precedents as SIH26107 apply architecturally (LawPal, LexRAG).

What would make this stand out

- The two-stage "classify regime, then retrieve" design — the single most valuable technical differentiator, since flat RAG will conflate patent-law answers with drug-classification answers on questions spanning both.
- A "which regime(s) apply to my formulation" guided intake, mirroring SIH26107's idea but essential here given the far greater breadth.
- Explicit **ABS-obligation flagging** as its own distinct output category — not folded into general legal text — since it's the ethically highest-stakes part of the PS and the easiest for a naive chatbot to miss entirely.

5 Clarity of the Problem Statement

What's asked: a multilingual, source-cited RAG assistant covering patents/GI/trademarks/copyright/designs/trade-secrets/plant-variety rights/ABS duties plus drug-regulatory classification, for practitioners, researchers, AYUSH startups, MSMEs and cultivators.

Where teams misread it: scope-creep denial. Seeing "six-plus legal regimes" and either (a) trying to shallowly cover all of them — confidently wrong across the board — or (b) missing that drug-regulatory classification is itself a distinct, hard sub-problem, not a footnote.

Framing that lands: openly state the 2-3 regime scope-down as a strategic choice ("depth over false breadth"), and make the regime-classification step *visible* in the demo UI rather than hidden — it shows judges you understood the problem's real difficulty.

6 Evaluator's Perspective

Given the source specificity the PS itself names (TKDL, India Code, IP India), expect an AYUSH-ministry-aligned or IP-law-literate evaluator to test the demo with a moderately complex, multi-regime scenario.

- **Matters most:** honesty about TKDL access limitations, citation discipline, correctness on the regime-classification step, and whether ABS/community-knowledge is addressed at all — dropping it is a visible, embarrassing gap given how central it is to the PS's own background.
- **Red flags:** claiming full TKDL integration without disclosing its restricted-access reality, a demo that only ever shows single-regime questions, zero mention of ABS obligations anywhere.

7 Team Fit & Execution Strategy

Skills needed: NLP/LLM engineer(s), backend/API developer, frontend developer, and — more than either other PS — a genuine legal/IP-domain researcher, because the entire value proposition rests on correctly representing six different legal regimes. This is not a role an AI coding assistant can substitute for.

Ideal 6-person ratio: 2 AI/NLP · 1-2 full-stack · 1-2 domain/legal-research (the largest domain-research allocation of the three PS) · 1 UX/presentation. Legal accuracy *is* the product here, not a nice-to-have.

1. Before any code: have the domain researcher(s) read the new AYUSH examination guidelines and decide which 2-3 regimes the team will cover, and why.
2. Build the regime-classification step as a standalone, testable component first — it's the architectural backbone.
3. Build retrieval within each chosen regime using India Code + IP India public sources.
4. Add ABS-obligation flagging as an explicit, separate output.
5. Multilingual support last.

8 AI-Buildability Split (20/80) 🤖

The fast 20%: the RAG plumbing (identical to SIH26107), chat UI, a basic multilingual translation wrapper.

The real 80%: correctly scoping which 2–3 of six-plus regimes to cover, and defending that choice; building regime-classification logic, which requires genuinely understanding how patent law, GI law and drug-regulatory classification differ — a subtle, judgment-heavy legal distinction, not a pattern-matchable one; handling TKDL's real access restrictions honestly instead of quietly working around them with scraped or invented data; and designing ABS-obligation flagging, which requires understanding biodiversity law well enough to know when it's even triggered.

Risk of leaning on AI alone: an AI-assistant-built version of this system will almost certainly conflate legal regimes — answering a patent question with GI-law citations, or vice versa — because it has no independent grip on these being legally distinct frameworks. In front of a legally-literate judge, this is the fastest way to lose credibility, and it's precisely the failure mode this PS was written to prevent.

On-the-spot test: "A cultivator supplied a plant-based formulation involving community traditional knowledge — what does your system say about their ABS obligations?" If ABS-flagging is real and not decorative, the team answers live. If it was never built as a distinct reasoning path, this question exposes that instantly — and it's exactly the kind of question this PS's panel is likely to ask, given how prominently ABS features in the PS's own text.

9 Data & Resource Availability

- **Public:** India Code (indiacode.nic.in) and IP India's databases (patents/InPASS, trademarks, designs, GI Registry) are genuinely public and usable.
- **Restricted:** TKDL (tkdl.res.in) — named in the PS's own suggested sources, but has historically operated under access-agreement-based sharing, primarily with patent offices, not open bulk access. A meaningful gap between what the PS suggests and what's actually obtainable.
- **If the ideal source is unavailable:** don't claim TKDL integration. Scope the "traditional knowledge" angle around what's independently documentable — published academic/ethnobotanical literature, the CNLU paper's own framing, publicly disclosed prior-art cases — and state plainly that full TKDL search needs an institutional access agreement the team doesn't have.
- **Backup plan:** corpus from India Code + IP India public data + the new AYUSH examination guidelines + a small, manually-curated set of illustrative TK/ABS case examples from public academic literature, clearly labeled illustrative rather than a live TKDL search.

10 Judge Q&A Stress-Test

Q1 · Have you actually integrated TKDL? Show me a live TKDL search.

ANSWER Be direct — TKDL operates under restricted, access-agreement-based sharing primarily with patent offices, not open public bulk access, so full integration isn't something a hackathon team can legitimately claim. The TK/ABS-awareness layer uses publicly documented academic case examples instead, and is architected to plug into TKDL given an institutional access agreement.

FOLLOW-UP *"So how does a real user get TK-search coverage today?" — point them to TKDL's own defensive-search channel via patent offices; position your tool as complementary guidance, not a TKDL replacement.*

Q2 · My formulation involves both a patent question and a drug-classification question. Does your system handle that, or just pick one?

ANSWER Walk through the regime-classification step live — the system identifies that the query spans two regimes and answers both, each with its own citations, rather than collapsing into a single answer.

Q3 · Why did you only cover 2-3 of the regimes the PS names, not all of them?

ANSWER State it as a deliberate choice — six-plus overlapping legal regimes at genuine accuracy isn't credible in a hackathon timeframe, and a system that's shallow across all of them is more dangerous than one honest about its scope, since users could rely on a wrong answer in an area that isn't actually covered well.

Q4 · What happens when a cultivator's community traditional knowledge is involved — does your system say anything about their rights?

ANSWER Show the explicit ABS-obligation-flagging output — its own distinct category, not folded silently into general legal text, because it's the highest-stakes, most easily-missed part of the PS.

Q5 · Is your legal information actually current? AYUSH IP guidelines have been changing recently.

ANSWER Reference the new draft AYUSH-invention examination guidelines from the Indian Patent Office directly in the corpus, and show a "source last updated" indicator on citations. Acknowledge that a production system needs an ongoing monitoring pipeline against Patent Office notifications.

 YELLOW LIGHT

The most genuinely novel of the three PS — no existing public tool does this — but full regime coverage and TKDL access are both harder to get than the PS text implies. The verdict rests entirely on whether the team scopes down honestly instead of quietly cutting corners on legal accuracy.

SOURCES REFERENCED IN THIS DOSSIER

[eMaap — National Legal Metrology Portal](#)

[Legal Metrology \(Packaged Commodities\) Amendment Rules 2026 — e-commerce origin disclosure](#)

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[BIS Care App — official page](#)

[What is the BIS Care App & how to use it](#)

[LawPal — RAG-based legal accessibility system for India \(arXiv\)](#)

[LexRAG — multi-turn legal-consultation RAG benchmark \(ACM SIGIR 2025\)](#)

[RAG chatbot for Air Force policy & logistics compliance](#)

[TKDL — Traditional Knowledge Digital Library \(CSIR\)](#)

[Draft guidelines for processing AYUSH patent applications](#)

[Protecting AYUSH As A Traditional Knowledge In The AI Era \(CNLU\)](#)

Analysis built from the three problem statements' official text (SIH 2026 dataset) plus the live web research linked above. Verdicts are directional judgement calls for hackathon strategy, not a guarantee of evaluator scoring — validate the sharpest claims (accuracy numbers, data access, regulatory dates) yourselves before quoting them on stage.